

Autocomplete

The agent's output reaches the open file and nothing else. A human accepts every edit with a keypress, and nothing else about the process changes. The fixed low extreme of the spectrum — and the only archetype whose defining property is architectural rather than configured.

Vocabulary as of 2026-08-31 Taxonomy: ADR-0004 Sources last verified 2026-09-02



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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01 What this archetype is

An Autocomplete ceiling permits model-generated text to reach **the open file, and nothing else** — and only through a human keypress. The surface is a decoration in an editor buffer: a candidate string is drawn as ghost text, Tab commits it, continued typing discards it. Nothing becomes an edit until the developer commits it while looking at it [• Tooling: Inline & Chat, Aug 2026](#).

Concretely, the step the human stops doing is **typing the characters — and nothing else**. The developer still decides what to build, still reads what appears, still decides whether it is right, and still runs it. Reading is not removed; it is moved earlier and made continuous. This archetype does markedly less than its reputation: the widely-quoted figures — 50% of code characters completed by AI at Google, *"an average of 46% of a developers' code"* from GitHub — describe *character provenance*, not delegated decisions [• Google, Jun 2024](#) [• GitHub, Feb 2023](#). No decision changes hands. That is why the owner fixed this as the spectrum's low extreme: inline AI edits, with nothing else about the process changing [• ADR-0004](#).

02 The boundary crossed

None — this is the fixed low extreme. The boundary that defines Autocomplete is the one **not** crossed: **propose versus act**. Across fourteen completion products surveyed, not one documents a completion engine that executes a command, writes to a file the developer is not editing, or makes any outbound call beyond the inference request itself. The architecture forecloses it: a completion engine has no tool surface, so there is nothing to grant [Tooling: Inline & Chat, Aug 2026](#).

That crossing point is one keypress wide and has a street address: VS Code's terminal inline chat offers a **Run (⌘Enter)** control that executes the suggested command. Everything on the completion side of that control proposes text; everything past it can act [VS Code docs, Aug 2026](#). And the two sides differ in kind, not degree: **completion acceptance cannot be batched or automated anywhere in the surveyed field** — no vendor ships an auto-accept for ghost text — while in-editor *edit* acceptance acquires an off switch the moment a team crosses over (VS Code's `chat.editing.autoAcceptDelay` converts diff review into a countdown timer) [Tooling: Inline & Chat, Aug 2026](#). Autocomplete is therefore the only archetype whose ceiling is enforced by architecture. Every archetype further along the spectrum holds its position by configuration.

03 What a human still gates

Everything. Every character enters the file through a keypress made while looking at the suggestion, and the unit of approval is often *smaller* than one suggestion — word-by-word and line-by-line partial acceptance is universal across the category [Tooling: Inline & Chat, Aug 2026](#).

What varies within the archetype is not whether the human gates, but **what the gating is worth**. GitHub's own researchers found that acceptance rate — the obvious measure of this archetype's central act — tracks how productive developers *feel*, not what persists in the code [Ziegler et al., 2022](#). And the one input-scoping control the category ships leaks: Copilot content exclusion suppresses suggestions in excluded files, yet *"Copilot may use semantic information from an excluded file if the information is provided by the IDE indirectly"* — and it stops working entirely in edit and agent modes [GitHub Docs, Aug 2026](#).

INDEPENDENCE RULE

Containment and human oversight move independently. Autocomplete pins containment at its minimum; it does not guarantee attentive oversight — §06 takes that up. The spectrum position above names only the reach.

04 Who works this way

The anchor exemplar is **Salvatore Sanfilippo (antirez)**, creator of Redis — with no commercial stake in any AI product, one of the cleanest evidence classes available. He uses frontier models daily and refuses agents entirely, and the withholding is the method: he moves code by hand between his terminal and a web chat interface — *"Always be part of the loop by moving code by hand from your terminal to the LLM web interface: this guarantees that you follow every process."* The friction is deliberate; it is what guarantees he reads every line. His objection to agents is a quality claim, not a productivity one: autonomous agents

produce "*fragile code bases that are larger than needed, complex, full of local minima choices*"

• antirez, mid-2025 . Note that his practice sits *below* even this archetype's ceiling — he denies the model the open file too — which illustrates that an archetype names what the settings permit, not what any individual does.

As a population, this end is measured worse than its reputation suggests. The only direct measurement of "inline only" found anywhere: 13.8% of Stack Overflow 2025 respondents prefer copilot/autocomplete mode exclusively • Stack Overflow 2025 . The headline adoption number this archetype is usually folded into is unusable as a point estimate: four credible instruments asking four different questions of four different populations produce **75–90%**, every one opt-in, none publishing a non-response analysis — quote the range and name the instrument, never a single figure • Evidence Base, Aug 2026 . (There is no Stack Overflow 2026 survey; anything citing one is citing mislabelled 2025 data.) The refusal axis puts this archetype in context: 16.2% of developers won't use AI tools at all, and 37.9% have no plans to use agents — abstainers and autocomplete-only developers together are a far larger population than the discourse implies • Practitioner Exemplars, Aug 2026 .

05 What it gains you, what it costs

WHAT IT GAINS YOU

- The only region of the spectrum with clean randomised evidence in its favour: three field RCTs across 4,867 developers found **+26.08% completed tasks** for inline completion — peer-reviewed, pre-registered, the strongest single piece of positive evidence in the field • Cui et al., Feb 2026 .
- No new verification machinery is demanded — the existing process, human included, is unchanged (§06).
- The only region where the egress question can be *eliminated* rather than managed: JetBrains Full Line completion "*runs entirely on your local device without sending any code over the internet*", and air-gapped deployments are shipped • JetBrains docs, Aug 2026 . Teams with a hard data-egress constraint have a real product choice here and progressively fewer further along the spectrum.

WHAT IT COSTS YOU

- The smallest delta on the spectrum: the human still does everything except type. Teams adopting "AI" at this ceiling and expecting transformation are buying character provenance (§01).
- Reading is made continuous — a per-keystroke stream of accept/reject micro-decisions — and the best-evidenced fact about that stream is that its acceptance rate measures satisfaction, not output • Ziegler et al., 2022 .
- For every cloud product, each keystroke can send local editing context to the provider — the category's one genuinely non-zero risk surface, and privacy modes govern training use, not transmission • Tooling: Inline & Chat, Aug 2026 .

06 What it demands of verification

The entailment — whatever a human stopped doing must be replaced by engineered verification — reaches its minimum here, because almost nothing human was removed. No completion product ships a test run, a lint gate, or a static-analysis check between suggestion and acceptance; the sole automated check found in the category is Copilot's public-code matching, and it is a licensing check, not a correctness one

● Tooling: Inline & Chat, Aug 2026 . The safety case is one mechanism: a human reads every line, at the moment it appears.

Two honest caveats keep that from being a comfortable sentence. First, **the mechanism is attention, and attention is the control this project's evidence trusts least**: the vendor holding the telemetry reports permission-prompt approval at ~93% rising to 97% and measures human review catching 13.6% of dangerous commands against 89% for automated checking ● Anthropic telemetry, 2026 . Those figures were measured on agent permission prompts, not ghost text — but they establish that "a human looked at it" degrades with volume, and this archetype runs the highest approval volume on the spectrum. antirez's hand-carried workflow is best read as engineering against exactly this: he adds friction to protect the reading (§04). Second, **what this archetype demands is the discipline of staying in it deliberately**. The category boundary is architectural (§02), but most completion products now ship inside editors that also carry edit and agent surfaces whose approval steps have off switches. A team whose archetype is Autocomplete holds that position by not crossing a line that is one keypress — or one settings value — away ● Tooling: Inline & Chat, Aug 2026 .

07 Governance implications

The least exposed position on the spectrum — and the exposure that remains is instructive because every archetype inherits it. The provider disclaims defect liability even here: *"You are responsible for reviewing, testing, and validating any Output before use"* ● GitHub ToS §J.4, 2026 . At this ceiling the team is actually doing the thing the terms assume, line by line — the disclaimer and the practice align, which is not true anywhere further along the spectrum.

Authorship is at its most defensible here: a human continuously selects, edits, and integrates suggestions at character granularity, retaining control over the expressive elements — the very thing the US Copyright Office's threshold turns on. Nothing in law or standards requires a human to read source code, so this archetype's discipline is a choice, not a compliance obligation; teams here should know they are exceeding every verified regime, not meeting one ● Governance, Legal & IP, Aug 2026 . The practical governance question that remains is data egress (§05) — a procurement and deployment question with a complete answer (the local lane), not a process one.

08 The posture that fits

This fits a team that believes reading every line is non-negotiable, and that anything reducing what a human reads costs more than it saves.

The safety belief underneath: *a human reads every line*. Uniquely on the spectrum, the architecture backs the belief — there is no configuration under which the reading moment disappears. What the architecture cannot supply is the reading itself: the belief is honest only while the attention is real, and the exemplar who holds this posture most publicly spends his engineering effort making the attention unavoidable rather than assuming it.

— Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, `ADR-0004` — the taxonomy contract: definitions, boundary criterion, the fixed low extreme. Adopted 2026-08-31.
- 2 **STUDY** This project, `research/tooling-inline-and-chat.md` — fourteen completion products and the in-editor edit surfaces, read from vendor documentation; the propose-vs-act boundary, acceptance mechanics, egress, and `chat.editing.autoAcceptDelay`. Closed 2026-08-30.
- 3 **ANECDOTE** Salvatore Sanfilippo, "Coding with LLMs in the summer of 2025 (an update)", antirez.com/news/154. Stake: none — independent, no product. Position dated mid-2025 and may have moved since. Jul 2025.
- 4 **STUDY** This project, `research/evidence-base.md` — the four adoption instruments and their traps; the no-2026-survey finding. Closed 2026-08-27.
- 5 **SURVEY** Stack Overflow Developer Survey 2025 (49,000+ respondents; per-question N ~26.5k–33.7k; opt-in) — 13.8% prefer copilot/autocomplete mode exclusively. Fielded May–Aug 2025.
- 6 **STUDY** This project, `research/practitioner-exemplars.md` — the refusal axis (16.2% / 37.9%, reconstructed with corrected denominators) and the antirez entry. Closed 2026-08-27.
- 7 **STUDY** Ziegler et al., "Productivity Assessment of Neural Code Completion", arXiv:2205.06537 — observational, vendor-authored (GitHub): acceptance rate drives *perceived* productivity. May 2022.
- 8 **VENDOR** Google Research blog, "AI in Software Engineering at Google" — 37% acceptance rate; 50% of code characters. Google measuring Google. 2024-06-06.
- 9 **VENDOR** GitHub blog — "an average of 46% of a developers' code" completed by Copilot. 2023-02-14.
- 10 **VENDOR** VS Code documentation, inline chat — the terminal Run (⌘Enter) control. Page dated 2026-08-26.
- 11 **VENDOR** GitHub Docs, "Content exclusion for GitHub Copilot" — indirect-context leak; unsupported in Edit and Agent modes. Read 2026-08-30.

- 12 **STUDY** Cui, Demirer, Jaffe, Musolff, Peng & Salz, *Management Science* — three field RCTs, N=4,867, GitHub Copilot inline completion, +26.08% completed tasks (SE 10.3%); pre-registered AEARCTR-0014530. Published 2026-02-27.
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- 13 **VENDOR** JetBrains documentation, Full Line Code Completion — fully local inference; Tabnine documents an air-gapped deployment. Read 2026-08-30.
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- 14 **VENDOR** Anthropic-published telemetry: ~93%→97% permission-prompt approval; human review catching 13.6% of dangerous commands vs 89% automated (n=1,053). Measured on agent prompts, not completion. 2026.
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- 15 **VENDOR** GitHub Terms of Service §J.4 (contract text). Effective 2026-04-27.
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- 16 **STUDY** This project, `research/governance-legal-ip.md` — no verified regime requires a human to read source code; the USCO control-over-expression threshold. Closed 2026-08-30, corrected 2026-08-31.
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